Reg. No. | | | | | | | | | | |

[12AE72]

SEVENTH SEMESTER B.E EXAMINATION, DECEMBER – 2015

AEROSPACE AVIONICS

Time: 3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full Questions.
2. Draw neat sketches wherever required.
3. Assume suitably any missing date and state the same.

1.	**a.**	Explain the operation of instrument landing system covering localizer and glide slope operate in ensuring smooth landing of an aircraft.	**07**
	b.	With the help of sketches of antenna pattern of localizer and glide slope explain how correct landing path is achieved.	**07**
2.	**a.**	Explain the operating principle of VOR as an important navigation guide.	**07**
	b.	What is distance measuring equipment? How does it function?	**07**
3.	**a.**	What is a primary radar and what are the information it provides on the target?	**05**
	b.	Explain what is a secondary radar and compare its performance with primary radars.	**05**
	c.	In general, list out all the performance parameters of the target that can be acquired by any of the radar.	**04**
4.	**a.**	Explain the principle behind measurement of temperature using thermal resistance element. How is the signal conditioning and calibration carried out?	**05**
	b.	What is a thermos-couple? Take examples of any two types and explain the operation and difference in sensitivity.	**05**
	c.	What is cold junction compensation and why is it required in aircraft temperature measurement?	**04**
5.	**a.**	Why are avionics systems integrated through special bus architecture?	**05**
	b.	Explain briefly the architecture of MIL-1553B bus used in avionics system.	**05**
	c.	Compare the major difference between MIL-1553B bus and ARINC429 system.	**04**
6.	**a.**	What is pitot-static system and its architecture?	**05**
	b.	How is this system used to provide display in cockpit of essential air data?	**04**
	c.	Explain the principle of operation of vertical speed (rate of climb).	**05**

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7. a. What are the major navigational systems installed in an aircraft? Briefly explain the operation of each one of them. **07**

b. What is WGS-85 Geodesy and its application? **07**

8. Write short notes on any four of the following. **14**

i. Mach number.

ii. Automatic direction finder.

iii. Inertial navigation system.

iv. Steps involved in computation of true air speed from indicated air speed.

v. GPS as a navigational aid.

vi. Pulse Doppler radar.